

The CD47 "don't eat me signal" is highly expressed in human ovarian cancer.

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The CD47 "don't eat me" signal allows tumor immune evasion. We tested the association of CD47 expression with outcomes in EOC. CD47 expression was examined within the TCGA database for ovarian carcinoma. For validation, IHC was performed on a TMA consisting of specimens from 265 patients with EOC. The medical records of the patients were also retrospectively reviewed to correlate demographic and survival data. CD47 was amplified in 15/316 (5%) ovarian serous cancers in TCGA. In the validation cohort, the majority of patients had stage III/IV disease (208/265, 78.4%). CD47 expression was seen in 210/265 (79.2%). Patients were categorized into CD47hi (129/265; 48.7%) versus CD47lo (136/265; 51.3%). Patients with CD47lo tumors were more likely to have a complete response to adjuvant therapy than CD47hi (65% vs 50%, $p=0.026$). Although there was a trend towards an increase in median OS (37.64 vs 45.26 months, $p=0.92$) in the CD47lo group compared with CD47hi, the difference was not significant. CD47 is expressed at high frequency in EOC. Patients with CD47lo EOC had a better treatment response to standard therapy, and trended towards improved OS. This demonstrates that while CD47 may be an immunologic shield that may be considered for targeted therapies, it is likely that it operates in concert with other mechanisms of immune evasion. Future studies to evaluate CD47 expression with other known mechanisms of immune escape in the tumor microenvironment may help further define its role.

Expression of CD47 antigen in Reed-Sternberg cells as a new potential biomarker for classical Hodgkin lymphoma.

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CD47 over expression has been reported in several tumor subtypes. CD47 interacts with SIRPalpha on macrophages inhibiting phagocytic signal, providing a survival advantage to tumor. CD47, therefore, represents a valuable target for immunotherapy and is currently under clinical investigation. We aimed to study CD47 expression in Hodgkin Reed Sternberg cells (HRS). We tested a polyclonal CD47 antibody (LifeSpan Biosciences, Seattle, WA) expression along with classical HRS cell markers on a tissue array of 16 classical Hodgkin Lymphoma (CHL) tumor biopsies obtained from newly diagnosed, non-selected patients (8 Female, 8 Male patients) in our institution from October 2016 to January 2018. Histologic subtypes were nodular sclerosis in 11 cases, mixed Cellularity in 3 cases and lymphocyte rich in 2 additional cases. Median age was 53 years (Range: 8, 74). Early stage disease was found in three patients without unfavorable prognostic factors according to EORTC and GHSG criteria, one patient with unfavorable prognostic factors and nine patients had advanced disease. Bulk disease was present in one patient. Normal lymphoid tissue and normal prostate epithelium were used as normal controls as recommended by manufacturer. Approval from the Local Ethical committee was obtained before any analysis. CD47 was overexpressed on all HRS cells with a characteristic dot-like pattern in 13/13 cases of CHL. HRS clearly expressed CD47 more intensely than infiltrating T and stromal cells. We propose that HRS cells, by up-regulating CD47, might avoid innate immunity check on tumor growth, which could be circumvented using blocking monoclonal antibodies.

CD47 is not Over-Expressed in Fibrolamellar Hepatocellular Carcinoma.

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CD47 is a transmembrane receptor that inhibits phagocytosis. Over-expression of CD47 is associated with an increased risk of tumor growth and metastasis. Clinical trials based on anti-CD47 therapy in adults are underway in a variety of malignancies. CD47 has been shown to be over-expressed in conventional hepatocellular carcinoma (HCC), a common liver tumor in adults. To our knowledge, there have been no studies to evaluate CD47 expression in the fibrolamellar subtype of HCC (FL-HCC), common in children and young adults. This study will evaluate CD47 expression in FL-HCC and shed light on its suitability for anti-CD47 therapy. Using immunohistochemistry, 10 samples of FL-HCC from 8 patients were evaluated for CD47 (anti-phagocytic) and calreticulin (pro-phagocytic) expression. By direct comparison, CD47 and calreticulin expression were evaluated in 21 samples of conventional HCC. Additionally, transcriptome sequencing to detect CD47 mRNA expression was performed on fresh tissue from 1 FL-HCC institutional patient and previously published sequencing data from 20 additional samples was reviewed. Immunohistochemistry showed only weak CD47 expression in 20% of FL-HCC samples. In contrast, 57% of conventional HCC samples showed CD47 expression. All (100%) FL-HCC samples showed moderate or strong calreticulin expression. The difference between CD47 and calreticulin expression in FL-HCC is statistically significant ($p=0.0007$). Transcriptome sequencing revealed no difference in CD47 expression between FL-HCC and normal liver samples. CD47 is not over-expressed in FL-HCC. Our studies provide no support for expanding ongoing clinical trials in adults to include children and young adults with FL-HCC.

Mononuclear phagocyte system blockade using extracellular vesicles modified with CD47 on membrane surface for myocardial infarction reperfusion injury treatment.

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Mesenchymal stem cell (MSC)-derived extracellular vesicles (EVs) with anti-apoptotic and anti-inflammatory properties have been intensively studied. However, rapid clearance by the mononuclear phagocyte system remains a huge barrier for the delivery of extracellular vesicle contents into target organs and restricts its wider application, particularly in the heart. CD47 is a transmembrane protein that enables cancer cells to evade clearance by macrophages through CD47- signal regulatory protein α binding, which initiates a "don't eat me" signal. This study aimed to explore the biodistribution and delivery efficiency of EVs carrying the membrane protein CD47 and specific anti-apoptotic miRNAs. EVs were isolated from MSCs overexpressing CD47 (CD47-EVs) and identified. Fluorescence-labeled EVs were injected through the tail vein and tracked using fluorescence imaging. In silico analysis was performed to determine miRNA profiles in MSCs and in a heart-derived H9c2 cardiomyoblast cell line under hypoxia vs. normoxia conditions. Electro CD47-EV was constructed by encapsulating purified CD47-EV with miR-21a via electroporation. The effect of miR21-EVs on the pro-apoptotic gene encoding phosphatase and tensin homolog (PTEN) was evaluated by dual-luciferase assay, qPCR, and western blotting. Exogenous miR21 distribution, PTEN protein level, blood vessel density, anti-apoptotic effect by TdT-mediated dUTP nick-end labeling staining, and macrophage and leukocyte infiltration in the myocardium were assessed by immunofluorescence staining. Cardiac functional recovery during the early stage and recovery period was evaluated using echocardiography. The results showed that CD47-EVs were still detectable in the plasma 120 min after the tail vein injection, compared to the detection time of less than 30 min observed with the unmodified EVs. More strikingly, CD47-EVs preferentially accumulated in the heart in the ischemia-reperfusion (I/R) + CD47-EV group [heart total fluorescence radiance ($\times 10^5$ Photons/sec/cm²/sr) 51.62 ± 11.30 v.s. 10.08 ± 3.15 in the

I/R + unmodified EVs group] 8 h post-injection. Exogenous miR-21 is efficiently internalized into cardiomyocytes, inhibits apoptosis, alleviates inflammation, and improves cardiac function. In conclusion, electro CD47-EVs efficiently improve biodistribution in the heart, shedding new light on the application of a two-step EV delivery method (CD47 genetic modification followed by therapeutic content electrotransfection) as a potential therapeutic tool for myocardial I/R injury that may benefit patients in the future.

CD47 in the Brain and Neurodegeneration: An Update on the Role in Neuroinflammatory Pathways.

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CD47 is a receptor belonging to the immunoglobulin (Ig) superfamily and broadly expressed on cell membranes. Through interactions with ligands such as SIRP α , TSP-1, integrins, and SH2-domain bearing protein tyrosine phosphatase substrate-1 (SHPS-1), CD47 regulates numerous functions like cell adhesion, proliferation, apoptosis, migration, homeostasis, and the immune system. In this aspect, previous research has shown that CD47 modulates phagocytosis via macrophages, the transmigration of neutrophils, and the activation of T-cells, dendritic cells, and B-cells. Moreover, several studies have reported the increased expression of the CD47 receptor in a variety of diseases, including acute lymphoblastic leukemia (ALL), chronic myeloid leukemia, non-Hodgkin's lymphoma (NHL), multiple myeloma (MM), bladder cancer, acute myeloid leukemia (AML), Gaucher disease, Multiple Sclerosis and stroke among others. The ubiquitous expression of the CD47 cell receptor on most resident cells of the CNS has previously been established through different methodologies. However, there is little information concerning its precise functions in the development of different neurodegenerative pathologies in the CNS. Consequently, further research pertaining to the specific functions and roles of CD47 and SIRP is required prior to its exploitation as a druggable approach for the targeting of various neurodegenerative diseases that affect the human population. The present review attempts to summarize the role of both CD47 and SIRP and their therapeutic potential in neurodegenerative disorders.

Increased lymphocyte activation and atherosclerosis in CD47-deficient mice.

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CD47, also known as integrin-associated protein (IAP), is a transmembrane protein with multiple biological functions including regulation of efferocytosis and leukocyte trafficking. In this study we investigated the effect of CD47-deficiency on atherosclerosis using a model of adeno-associated virus (AAV)-induced hypercholesterolemia. We observed increased plaque formation in CD47 null mice compared to wild-type controls. Loss of CD47 caused activation of dendritic cells, T cells and natural killer (NK) cells, indicating an important role for CD47 in regulating immunity. In particular, Cd47 deficiency increased the proportion of IFN- γ producing CD90+ NK cells. Treatment with depleting anti-NK1.1 monoclonal antibody (mAb), but not depleting anti-CD4/CD8 mAbs, equalized atherosclerotic burden, suggesting NK cells were involved in the enhanced disease in Cd47 deficient mice. Additional studies revealed that levels of CD90+ and IFN- γ + NK cells were expanded in atherosclerotic aorta and that CD90+ NK cells produce more IFN- γ than CD90- NK cells. Finally, we demonstrate that anti-CD47 (MIAP410) causes splenomegaly and activation of DCs and T cells, without affecting NK cell activation. In

summary, we demonstrate that loss of CD47 causes increased lymphocyte activation that results in increased atherosclerosis.

CD47: a new target in cardiovascular therapy.

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CD47, originally named integrin-associated protein, is a receptor for thrombospondin-1. A number of important roles for CD47 have been defined in regulating the migration, proliferation, and survival of vascular cells, and in regulation of innate and adaptive immunity. The recent discovery that thrombospondin-1 acts via CD47 to inhibit nitric oxide signaling throughout the vascular system has given new importance and perhaps a unifying mechanism of action to these enigmatic proteins. Here we trace the development of this exciting new paradigm for CD47 function in vascular physiology.

Target cell CD47 regulates macrophage activation and erythrophagocytosis.

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The ubiquitously expressed cell surface glycoprotein CD47 (integrin-associated protein, IAP) was originally identified as a regulator of integrin-dependent leukocyte responses to extracellular matrix proteins. However, it has been shown that CD47 has several important functions in addition to regulating integrin activation. Extensive studies in murine systems have shown that CD47 on erythrocytes and other cells can function as a regulator of target cell phagocytosis, by binding to the inhibitory receptor SIRPalpha on macrophages. In this way, macrophages are less likely to phagocytose an autoimmune sensitized cell with CD47 on its surface than a CD47-deficient cell where this inhibitory mechanism will not be engaged. The CD47-SIRPalpha interaction seems to be important in limiting destruction of host cells in experimental models of autoimmune diseases like autoimmune hemolytic anemia (AIHA) or immune thrombocytopenia, where macrophages destroy antibody or complement opsonized cells.

Integrin-associated protein (CD47) and its ligands.

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Integrin-associated protein (IAP or CD47) is a receptor for thrombospondin family members, a ligand for the transmembrane signaling protein SIRP alpha and a component of a supramolecular complex containing specific integrins, heterotrimeric G proteins and cholesterol. Peptides containing a VVM motif in the C-terminal domain of thrombospondins are agonists for CD47, initiating heterotrimeric Gi protein signaling that augments the functions of integrins of the beta 1, beta 2 and beta 3 families, thus modulating a range of cell activities including platelet activation, cell motility and adhesion, and leukocyte adhesion, migration and phagocytosis.

[CD47 and leukemia stem cells].

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CD47, also known as integrin-associated protein (IAP), is an immunoglobulin-like protein. It can inhibit the phagocytosis of macrophages through binding with signal-regulatory protein alpha chain of inhibitory receptor on macrophage (SIRP α). The expression of CD47 on normal hematopoietic stem cells (HSCs) is useful for maintaining the stability of HSCs in body, but the high expression of CD47 existed on leukemia stem cell (LSCs) of AML patients which can reduce the macrophage-induced phagocytosis to LSCs and decrease the clearance of innate immune system of organism to LSCs. In this article, the expression and function of CD47 on HSCs and LSCs as well as the role of CD47 in the prognosis and target therapy of AML are reviewed.
